

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester IV)
EnterPrise Java
Practical-VIII

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8. Implement the following EJB applications with different types of Beans

A. Develop simple EJB application to demonstrate Servlet Hit count using Singleton

Session Beans.

Code:-

Code(index.jsp):-

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    <title>Hit Count Demo</title>
  </head>
  <body>
    <form action="ShowHitCountServlet" method="post">
      <input type="submit"
        value="Take Hit Count demo with Session Singleton Bean"/>
      <h1>Nandini Chaudhary</h1>
    </form>
  </body>
</html>
```

Code(STHitCntBean.java):-

```
package EJPracts;
import javax.ejb.Singleton;
@Singleton
public class STHitCntBean {
    private int count;
    public synchronized int inceHitCount() {
        return count++;
    }
}
```

Code(CCClient.java):-

```
package servalet;
import EJPracts.STHitCntBean;
import java.io.IOException;
import java.io.PrintWriter;
import javax.ejb.EJB;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/ShowHitCountServlet")
```

```

public class CCClient extends HttpServlet {
    @EJB
    private STHitCntBean cntBean;
    protected void processRequest(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Hit Count</title>");
            out.println("</head>");
            out.println("<body>");
            int count = cntBean.getHitCount();
            out.println("<h1>You have visited this page: "
                + count + " time/s</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }
    @Override
    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        processRequest(request, response);
    }
}

```

Output:-



B.Develop a simple visitor Statistics application using Message Driven Bean [MDB].

Code(index.jsp):-

```

<%@page import="javax.jms.TextMessage"%>
<%@page import="javax.jms.JMSException,
    javax.naming.InitialContext,
    javax.jms.MessageProducer,
    javax.jms.Session,
    javax.jms.Connection,
    javax.jms.Queue,
    javax.jms.ConnectionFactory,
    javax.jms.TextMessage" %>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<%!
    private static ConnectionFactory connFact;
    private static Queue queue;

```

```

    Connection conn = null;
    Session mySess = null;
    MessageProducer msgProducer = null;
    TextMessage msg = null;
%>
<html>
<head>
    <meta http-equiv="Content-Type"
        content="text/html; charset=UTF-8">
    <title>Message Driven Bean</title>
</head>
<body style="background-color:gray;text;">
<h1>Welcome To Message Driven Bean</h1>
<%
    try {
        InitialContext ctx = new InitialContext();
        queue = (Queue) ctx.lookup("jms/Queue");
        connFact = (ConnectionFactory)
            ctx.lookup("jms/QueueFactory");
        conn = connFact.createConnection();
        mySess = conn.createSession(
            false,
            Session.AUTO_ACKNOWLEDGE
        );
        msgProducer = mySess.createProducer(queue);
        msg = mySess.createTextMessage();
        msg.setText(request.getRemoteAddr());
        msgProducer.send(msg);
        conn.close();
    } catch (JMSEException e) {
        System.out.println(
            "Exception occurred: " + e.toString()
        );
    } catch (Exception e) {
        System.out.println(
            "Exception occurred: " + e.toString()
        );
    }
%>
</body>
</html>

```

Code(visitorStatMDB.java):-

```

package EJPracts;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import javax.ejb.ActivationConfigProperty;
import javax.ejb.MessageDriven;
import javax.jms.JMSEException;
import javax.jms.Message;

```

```

import javax.jms.MessageListener;
import javax.jms.TextMessage;
@MessageDriven(
    activationConfig = {
        @ActivationConfigProperty(
            propertyName = "destinationType",
            propertyValue = "javax.jms.Queue"
        ),
        @ActivationConfigProperty(
            propertyName = "destinationLookup",
            propertyValue = "jms/Queue"
        )
    }
)
public class VisitorStatMDB implements MessageListener {
    @Override
    public void onMessage(Message message) {
        try {
            if (message instanceof TextMessage) {
                String host =
                    ((TextMessage) message).getText();
                System.out.println("Visitor Host: " + host);
                // MySQL Driver
                Class.forName("com.mysql.jdbc.Driver");
                // Database connection
                Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/reg",
                    "root",
                    "root"
                );
                // Check whether host already exists
                String selectSQL =
                    "SELECT visitcount FROM visitorstat WHERE host=?";
                PreparedStatement ps =
                    con.prepareStatement(selectSQL);
                ps.setString(1, host);
                ResultSet rs = ps.executeQuery();
                if (rs.next()) {
                    // Existing visitor
                    String updateSQL =
                        "UPDATE visitorstat SET "
                        + "visitdate=NOW(), "
                        + "visitcount=visitcount+1 "
                        + "WHERE host=?";
                    PreparedStatement updatePS =
                        con.prepareStatement(updateSQL);
                    updatePS.setString(1, host);
                    updatePS.executeUpdate();
                    updatePS.close();
                } else {
                    // New visitor

```

```

        String insertSQL =
            "INSERT INTO visitorstat "
            + "(visitdate, host, visitcount) "
            + "VALUES (NOW(), ?, 1)";
        PreparedStatement insertPS =
            con.prepareStatement(insertSQL);
        insertPS.setString(1, host);
        insertPS.executeUpdate();
        insertPS.close();
    }
    rs.close();
    ps.close();
    con.close();
}
} catch (JMSEException e) {
    System.out.println(
        "JMS Exception: " + e.getMessage()
    );
} catch (Exception e) {
    System.out.println(
        "Database Exception: " + e.getMessage()
    );
}
}
}
}

```

Code(VStatelessBean.java):-

```

package EJPracts;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import javax.annotation.PostConstruct;
import javax.annotation.PreDestroy;
import javax.ejb.Stateless;
@Stateless
public class VStatelessBean {
    private Connection conn = null;
    private ResultSet rs = null;
    private Statement stmt = null;
    private String query = null;
    // This method runs automatically when Bean is created
    @PostConstruct
    public void connect() {
        try {
            // MySQL Connector 5.1 driver
            Class.forName("com.mysql.jdbc.Driver");
            // Database connection
            String connStr =
                "jdbc:mysql://localhost:3306/reg";
            conn = DriverManager.getConnection(

```

```

        connStr,
        "root",
        "root"
    );
    System.out.println("Connection Success");
} catch (ClassNotFoundException | SQLException e) {
    System.err.println(
        "Connection Failed: " + e.getMessage()
    );
}
}
}
@PreDestroy
public void disconnect() {
    try {
        if (conn != null) {
            conn.close();
        }
        System.out.println("Connection closed");

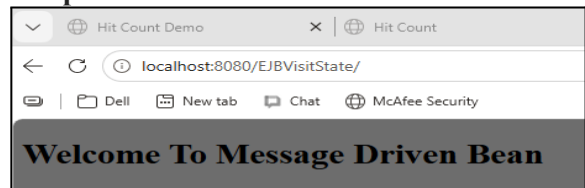
    } catch (SQLException e) {

        System.err.println(
            "Not able to close connection: "
            + e.getMessage()
        );
    }
}

public void addVisitor(String host) {
    try {
        stmt = conn.createStatement();
        query = "INSERT INTO visitorstat (host, visitcount) "
            + "VALUES (" + host + ", 1)";
        stmt.executeUpdate(query);
        System.out.println("New visitor added");
    } catch (SQLException e) {
        try {
            stmt = conn.createStatement();
            query = "UPDATE visitorstat "
                + "SET visitcount = visitcount + 1 "
                + "WHERE host = " + host + """;
            stmt.executeUpdate(query);
            System.out.println("Visitor count updated");
        } catch (SQLException ex) {
            System.err.println(
                "Cannot update visitor: "
                + ex.getMessage()
            );
        }
    }
}
}
}
}

```

Output:-



```

mysql> SELECT * From visitorstat;
+-----+-----+-----+
| visitdate | host | visitcount |
+-----+-----+-----+
| 2026-08-29 12:25:27 | 0:0:0:0:0:0:1 | 1 |
+-----+-----+-----+
1 row in set (0.00 sec)

```

C.Develop simple Marks Entry Application to demonstrate accessing Database using EJB.

Code(index.jsp):-

```
<%@page import="java.util.ArrayList,
    java.util.Iterator,
    java.util.List,
    javax.naming.InitialContext,
    EJPracts.MarksEntryBean"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<%!
    private static MarksEntryBean marksEntry;
    public void init() {
        try {
            InitialContext ic = new InitialContext();
            marksEntry = (MarksEntryBean) ic.lookup(
                "java:global/EJBMarksEntry/MarksEntryBean"
            );
        } catch (Exception ex) {
            System.out.println(
                "Database Error " + ex.getMessage()
            );
        }
    }
%>
<%
    if (request.getParameter("insertMarks") != null) {
        String name, mark1, mark2, mark3;
        name = request.getParameter("stuName");
        mark1 = request.getParameter("sub1");
        mark2 = request.getParameter("sub2");
        mark3 = request.getParameter("sub3");
        marksEntry.addMarks(
            name,
            mark1,
            mark2,
            mark3
        );
    }
%>
<html>
<head>
    <meta http-equiv="Content-Type"
        content="text/html; charset=UTF-8">
    <title>Marks Entry</title>
</head>
<body>
<h2 style="text-align:center;">
    Enter Details
</h2>
<br>
<form name="result" method="post">
    <table>
        <tr>
            <td>Enter Students Name</td>
            <td>
                <input type="text"
                    name="stuName">
            </td>
        </tr>
    </table>
</form>
</body>
</html>
```

```

        value="">
    </td>
</tr>
<tr>
    <td>Enter Subject 1 Marks</td>
    <td>
        <input type="text"
            name="sub1"
            value="">
    </td>
</tr>
<tr>
    <td>Enter Subject 2 Marks</td>
    <td>
        <input type="text"
            name="sub2"
            value="">
    </td>
</tr>
<tr>
    <td>Enter Subject 3 Marks</td>
    <td>
        <input type="text"
            name="sub3"
            value="">
    </td>
</tr>
<tr>
    <td colspan="2">
        <input type="submit"
            value="insertMarks"
            name="insertMarks">
    </td>
</tr>
</table>
</form>
</body>
</html>

```

MarksEntryBean.java

```

package EJPracts;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.Statement;
import javax.annotation.PostConstruct;
import javax.annotation.PreDestroy;
import javax.ejb.Stateful;
@Stateful
public class MarksEntryBean {
    private Connection conn = null;
    private Statement stmt = null;
    // Database connection
    @PostConstruct
    public void connect() {
        try {
            Class.forName("com.mysql.jdbc.Driver");
            String connStr =
                "jdbc:mysql://localhost:3306/reg";

```

Output:-

```

MySQL localhost:33060+ ssl reg SQL > DESC marks;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id     | int  | NO   | PRI | NULL    | auto_increment |
| stuName | varchar(45) | YES |     | NULL    |               |
| sub1   | varchar(20) | YES |     | NULL    |               |
| sub2   | varchar(20) | YES |     | NULL    |               |
| sub3   | varchar(20) | YES |     | NULL    |               |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.0067 sec)

```

```

Query OK, 0 rows affected (0.0251 sec)
MySQL localhost:33060+ ssl reg SQL > SHOW TABLES;
+-----+-----+
| Tables_in_reg |
+-----+-----+
| marks         |
+-----+-----+
1 row in set (0.0050 sec)

```

```

        conn = DriverManager.getConnection(
            connStr,
            "root",
            "Root@2007"
        );
        System.out.println("Connection Success");
    } catch (ClassNotFoundException | SQLException e) {
        System.err.println(
            "Connection Failed: " + e.getMessage()
        );
    }
}
// Close database connection
@PreDestroy
public void disconnect() {
    try {
        if (conn != null) {
            conn.close();
        }
        System.out.println("Connection closed");
    } catch (SQLException e) {
        System.err.println(
            "Not able to close connection: "
            + e.getMessage()
        );
    }
}
// Add student marks
public void addMarks(
    String name,
    String mark1,
    String mark2,
    String mark3) {
    try {
        stmt = conn.createStatement();
        String query =
            "INSERT INTO marks "
            + "(stuName, sub1, sub2, sub3) "
            + "VALUES ("
            + name + ","
            + mark1 + ","
            + mark2 + ","
            + mark3 + ")";
        stmt.executeUpdate(query);
        System.out.println("Marks inserted successfully");
    } catch (SQLException e) {
        System.err.println(
            "Error inserting marks: "
            + e.getMessage()
        );
    }
}
}
}

```